Experiment: 10

5G Network Slicing Environment Using MATLAB

Objective 1:

Analyze the creation and allocation of multiple network slices in a 5G network using MATLAB. Observe how different slices can be configured to support diverse service requirements and applications.

Objective 2:

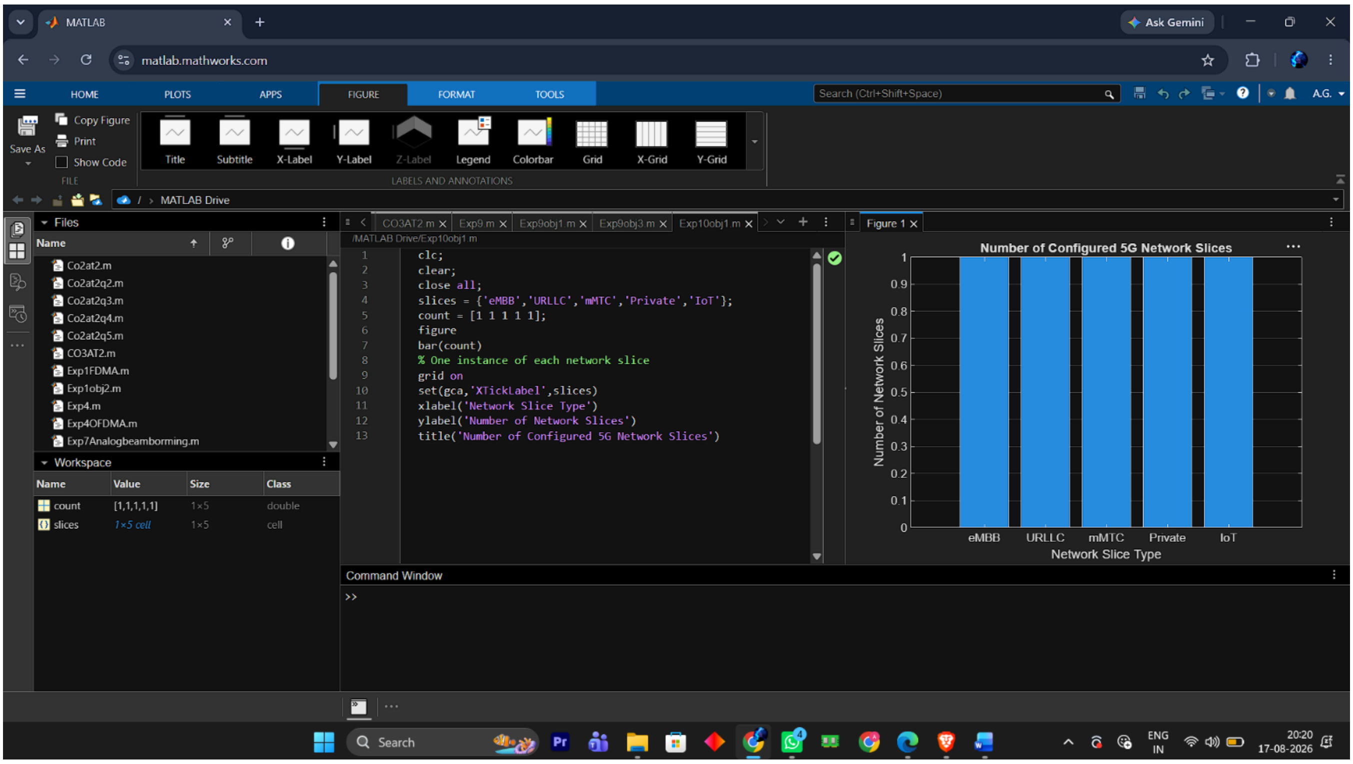
Compare the utilization of different network slices using MATLAB. Analyze how efficiently network resources are allocated and utilized across multiple slices.

Objective 3:

Analyze the end-to-end latency of different network slices using MATLAB. Compare the latency values to understand the Quality of Service (QoS) requirements of various 5G applications.

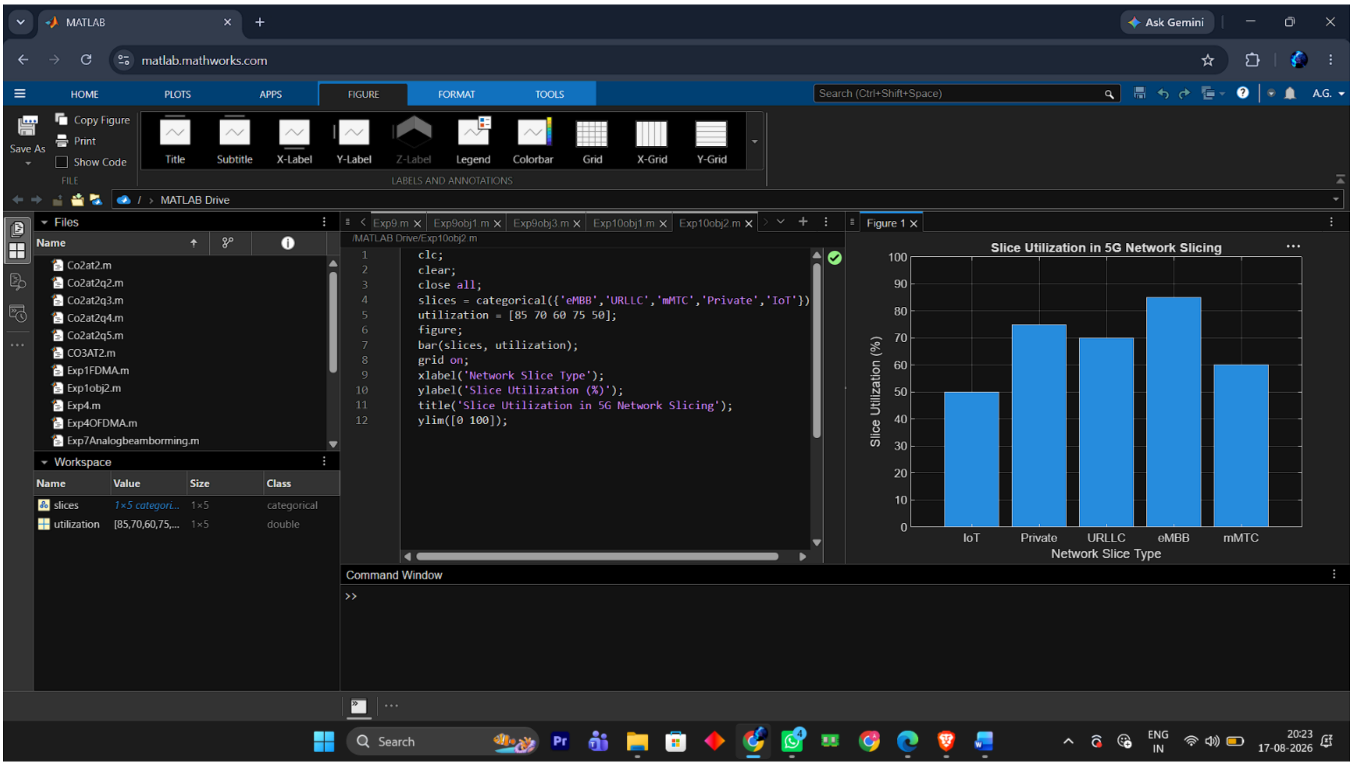
Objective 1 MATLAB Code and Output:

```
clc;
clear;
close all;
slices = {'eMBB', 'URLLC', 'mMTC', 'Private', 'IoT'};
count = [1 1 1 1 1];
figure
bar(count)
grid on
set (gca, 'XTickLabel', slices)
xlabel ('Network Slice Type')
ylabel ('Number of Network Slices')
title ('Number of Configured 5G Network Slices')
```



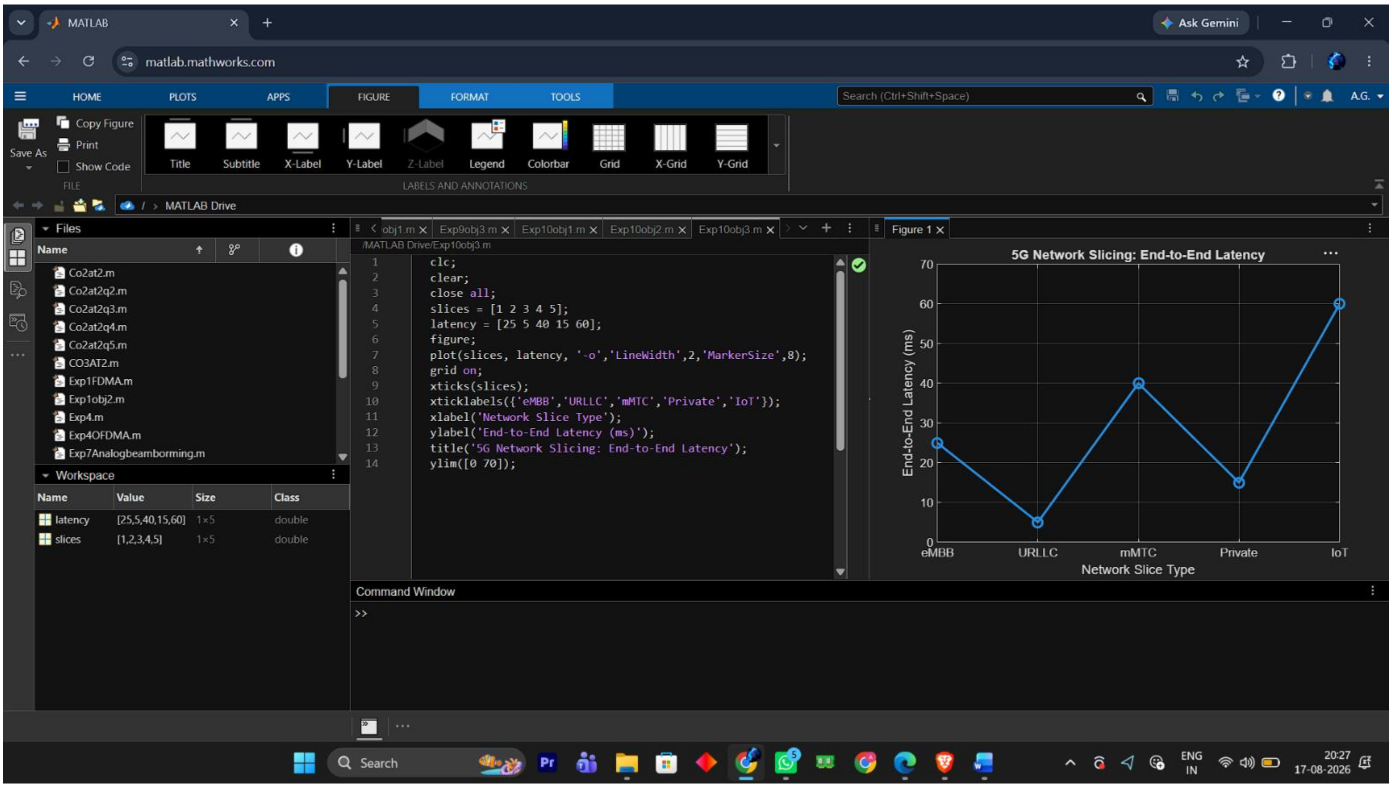
Objective 2: MATLAB Code and Output:

```
clc;
clear;
close all;
slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});
utilization = [85 70 60 75 50];
figure;
bar(slices, utilization);
grid on;
xlabel ('Network Slice Type');
ylabel ('Slice Utilization (%)');
title ('Slice Utilization in 5G Network Slicing');
ylim([0 100]);
```



Objective 3 MATLAB Code and Output:

```
clc;
clear;
close all;
slices = [1 2 3 4 5];
latency = [25 5 40 15 60];
figure;
plot (slices, latency, '-o', 'LineWidth', 2, 'MarkerSize', 8);
grid on;
xticks (slices);
xticklabels ({'eMBB','URLLC','mMTC','Private','IoT'});
xlabel ('Network Slice Type');
ylabel ('End-to-End Latency (ms)');
title ('5G Network Slicing: End-to-End Latency');
ylim([0 70]);
```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LABELS AND ANNOTATIONS
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Files
Co2at2.m
Co2at2q2.m
Co2at2q3.m
Co2at2q4.m
Co2at2q5.m
CO3AT2.m
Exp1FDMA.m
Exp1obj2.m
Exp4.m
Exp4OFDMA.m
Exp7Analogbeamborming.m
Workspace
Name Value Size Class
latency [25,5,40,15,60] 1×5 double
slices [1,2,3,4,5] 1×5 double
Exp9obj3.m
Exp10obj1.m
Exp10obj2.m
Exp10obj3.m
/MATLAB Drive/Exp10obj3.m
clc;
clear;
close all;
slices = [1 2 3 4 5];
latency = [25 5 40 15 60];
figure;
plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);
grid on;
xticks(slices);
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});
xlabel('Network Slice Type');
ylabel('End-to-End Latency (ms)');
title('5G Network Slicing: End-to-End Latency');
ylim([0 70]);
Figure 1
5G Network Slicing: End-to-End Latency
End-to-End Latency (ms)
Network Slice Type
eMBB
URLLC
mMTC
Private
IoT
Command Window
>>
Search
ENG
IN
20:27
17-08-2026